total amount of green house gases produced by human activities such as agriculture, industries, deforestation, waste disposal, buring fossil fuels directly or indirectly. It can be measured for an individual, family, organisation like industries, state level or national level. It is usually estimated and expressed in equivalent tons of CO_2 per year. The burning of fossil fuels releases $CO_2$and other green house gases. In turn these emissions trap solar energy and thus increase the global temperature resulting in ice melting, submerging of low lying areas and inbalance in nature like cyclones, tsunamis and extreme weather conditions. To reduce the carbon foot print we can follow some practices like (i) Eating indigenous fruits and products (ii) Reducing use of electronic devices (iii) Reduce travelling (iv) Avoid buying fast and preserved, processed, packed foods. (v) Plant a garden (vi) Reducing consumption of meat and sea food. Poultry requires little space, nutrients and less pollution compared cattle farming. (vii) reducing use of Laptops (when used for 8 hours, it releases nearly 2 kg. of CO_2 annually) (viii) Line drying clothes. (Example: If you buy imported fruit like kiwi, indirectly it increases CFP. How? The fruit has travelled a long distance in shipping or airliner thus emitting tons of CO_2)

Biochar

Biochar is another long term method to store carbon. To increase plants ability to store more carbon, plants are partly burnt such as crop waste, waste woods to become carbon rich slow decomposing substances of material called Biochar. It is a kind of charcoal used as a soil amendment. Biochar is a stable solid, rich in carbon and can endure in soil for thousands of years. Like most charcoal, biochar is made from biomass via pyrolysis. (Heating biomas in low oxygen environment) which arrests wood from complete burning. Biochar thus has the potential to help mitigate climate change via carbon sequestration. Independently, biochar when added to soil can increase soil fertility of acidic soils, increase agricultural productivity, and provide protection against some foliar and soil borne diseases. It is a good method of preventing waste woods and logs from getting decayed and instead we can convert them into biochar thus converting them to carbon storage material.

8.9 Rain water harvesting – RWH (Solution to water crisis – A ecological problem)

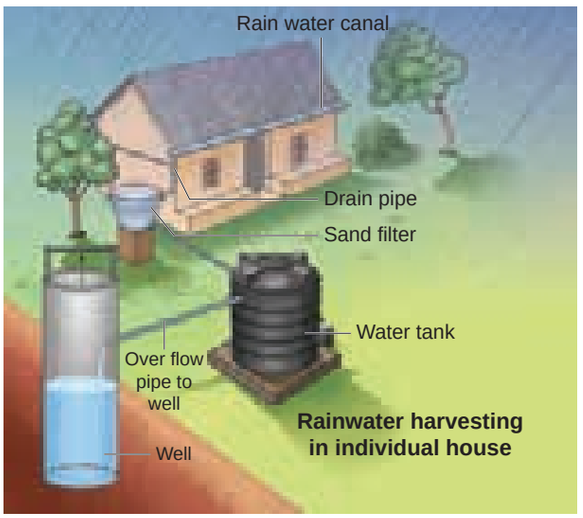


Figure 8.13: Pictures of Rain Water Harvesting Structures in Ooraniers

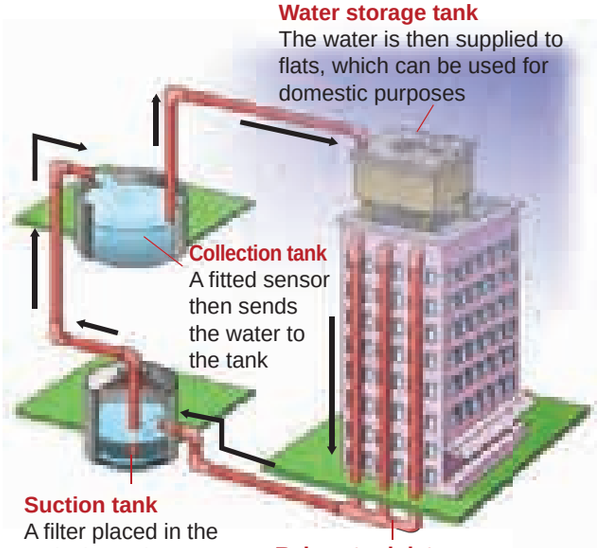


Figure 8.14: Rain Water Harvesting Structures in Water Supply sources

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